

Operationalising ICE-I: addressing equity, systems barriers, and artificial intelligence integration

We read with great interest the Viewpoint by Maïke Winters and colleagues,¹ published in *The Lancet Primary Care*, on health-care providers' role in patients' health-information journeys. The authors' innovative information space approach using the planet patient metaphor to conceptualise the complex dynamics among patients, health-care providers, and the modern information ecosystem is both compelling and timely. Their views on trust building through continuity of care and the proposed ideas, concerns, expectations, and information sources (ICE-I) framework are valuable contributions to understanding how health-care providers can navigate the challenges of health misinformation. I would like to offer some complementary observations to enrich this important discussion.

The framework would benefit from greater consideration of health equity and clinically vulnerable populations. Although the authors acknowledge how the information environment can disproportionately affect marginalised communities, further exploration of how information space dynamics differentially affect various demographic groups would strengthen the discussion. Rural populations face compound disadvantages, such as reduced access to the internet, as has been observed in the USA,² while racial and ethnic minorities encounter systematically different trust patterns and misinformation exposure, including disparities in trust towards health information sources and differential exposure to health misinformation through social and broadcast media.³ The ICE-I approach could be enriched with more specific guidance on adapting these conversations for diverse cultural contexts, varying literacy levels,

and different socioeconomic circumstances, to avoid exacerbating health disparities.

Although Winters and colleagues appropriately emphasise trust building and continuity of care, their discussion could benefit from giving greater attention to structural barriers. Current health-care systems often prioritise efficiency over relationship building, with time constraints, administrative burdens, and productivity pressures creating barriers to deeper patient engagement.³ A pilot study by Wood and colleagues⁴ found that 94.2% of health-care practitioners report encountering patient misinformation within the previous year, yet most had not received relevant training in misinformation management. Implementation of the ICE-I framework might be enhanced by acknowledging these system-level challenges and considering how health-care providers can develop the needed communication skills and information literacy competencies within existing constraints.

The discussion could be further enriched by considering the rapidly evolving technological landscape. A 2021 study reported on artificial intelligence (AI) detection systems achieving accuracy rates ranging from 77% to more than 85% in identifying health misinformation, although counter evidence often remains unavailable for novel false claims.⁵ Health-care systems might benefit from integrating technological solutions, including decision-support tools, real-time misinformation alerts, and AI-assisted patient education resources.

I appreciate the authors' important contribution to this crucial area and look forward to seeing how this framework develops as it is applied and refined in diverse health-care settings.

I declare no competing interests.

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- 1 Winters M, Wilhelm E, Pick J, et al. Navigating information space: the role of health-care providers in the health-information journey of patients. *Lancet Prim Care* 2025; **1**: 100040.
- 2 Tomascik J. Focus on agriculture: rural Americans need better broadband access. March 1, 2023. <https://www.agrinews-pubs.com/opinion/columnists/2023/03/02/focus-on-agriculture-rural-americans-need-better-broadband-access/> (accessed Nov 21, 2025).
- 3 Viswanath K, Ackerson LK. Race, ethnicity, language, social class, and health communication inequalities: a nationally-representative cross-sectional study. *PLoS One* 2011; **6**: e14550.
- 4 Wood JL, Lee GY, Stinnett SS, Southwell BG. A pilot study of medical misinformation perceptions and training among practitioners in North Carolina (USA). *Inquiry* 2021; **58**: 469580211035742.
- 5 Nasir JA, Khan OS, Varlamis I. Fake news detection: a hybrid CNN-RNN based deep learning approach. *Int J Inf Manag Data Insights* 2021; **1**: 100007.



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